

Web Programming Lab

Lab 5

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Batch A2

Homework:

Q1. Develop a simple web page to reproduce the given Captcha. Upon match, a suitable message has to be displayed. If there is a mismatch for more than 3 times, TextBox has to be disabled.

Backend:

```
def generate_captcha(length=5):
    return ''.join(random.choices(string.ascii_uppercase +
string.digits, k=length))

def captcha_view(request):
    result = None
    disabled = False

    if request.method == 'GET':
        request.session['captcha_attempts'] = 0
        request.session['captcha_value'] = generate_captcha()

    attempts = request.session.get('captcha_attempts', 0)

    if request.method == 'POST':
        if attempts >= 3:
            disabled = True
```

```

        result = "Too many failed attempts. CAPTCHA disabled."
    else:
        user_input = request.POST.get('captcha_input',
''').strip().upper()
        captcha_value = request.session.get('captcha_value')

        if user_input == captcha_value:
            result = "CAPTCHA validation successful!"
            request.session['captcha_attempts'] = 0
            request.session['captcha_value'] = generate_captcha()
        else:
            attempts += 1
            request.session['captcha_attempts'] = attempts
            request.session['captcha_value'] = generate_captcha()
            result = "CAPTCHA validation failed."

    if attempts >= 3:
        disabled = True

    return render(request, 'q3_t.html', {
        'result': result,
        'captcha': request.session['captcha_value'],
        'disabled': disabled,
        'attempts': attempts
    })

```

Template:

```

<!DOCTYPE html>
<html>
<head>
    <title>Captcha Page</title>
    <style>
        body {
            font-family: Arial;
            margin: 40px;
        }
        .captcha-box {
            font-size: 24px;
            font-weight: bold;
            letter-spacing: 4px;
            background: #f0f0f0;
            display: inline-block;

```

```

        padding: 10px 20px;
    }
    input[disabled] {
        background-color: #ddd;
    }
</style>
</head>
<nav class="navbar navbar-expand-lg navbar-dark bg-dark mb-4">
    <div class="container">
        <a class="navbar-brand" href="#">Lab 5</a>
        <div class="navbar-nav">
            <a class="nav-link" href="/lab5/q1/">Q1: Student Form</a>
            <a class="nav-link" href="/lab5/q2/">Q2: Promotion Check</a>
            <a class="nav-link" href="/lab5/q3/">Q3: Captcha Validation</a>
        </div>
    </div>
</nav>
<body>
<h2>Captcha Verification</h2>

<div class="captcha-box">
    {{ captcha }}
</div>

<form method="post">
    {% csrf_token %}
    <br><br>
    <input type="text" name="captcha_input"
        {% if disabled %}disabled{% endif %}
        required>
    <br><br>
    <button type="submit" {% if disabled %}disabled{% endif %}>
        Submit
    </button>
</form>

{% if result %}
    <p>{{ result }}</p>
{% endif %}

<p>Attempts: {{ attempts }}/3</p>

</body>
</html>

```

Output:

[Lab 5](#)

[Q1: Student Form](#) [Q2: Promotion Check](#) [Q3: Captcha Validation](#)

Captcha Verification

W L M L E

Submit

CAPTCHA validation failed.

Attempts: 3/3